

Merging and Pruning Cardiac MRI Features for Tsetlin Machine Classification

Lobna Ait Tghjdin

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Abstract

A Tsetlin Machine (TM) trained on 57 clinical cardiac features classifies the five ACDC diagnostic groups with $90.8\% \pm 2.7\%$ test accuracy. Four published ACDC feature sets were merged (656 features) and reduced to 313 features by physiology-driven rules. Adding radiomic texture and shape features lowered test accuracy to about 84%, even after redundancy removal.

1 Introduction

The aim of this work is an automatic, *interpretable* diagnosis of cardiac pathology from cine-MRI segmentations. Deep networks reach high accuracy on this task but give no readable justification. A Tsetlin Machine learns propositional rules (AND-clauses over thresholded features), so every prediction can be traced back to statements such as “LV ejection fraction < 0.28 AND maximal ES wall thickness < 21.6 mm”.

The work had three parts:

1. Merge the hand-crafted features published by four ACDC challenge teams (Wolterink, Khened, Isensee, Cetin) into one table.
2. Remove features that carry the same information, first by physiological reasoning, then by a correlation check on training data only.
3. Train and evaluate a multi-class Tsetlin Machine with a leakage-free cross-validation protocol, and compare three feature sets (reduced, clinical-only, all).

2 Dataset

The Automated Cardiac Diagnosis Challenge (ACDC, MICCAI 2017) dataset holds short-axis cine-MRI of 150 patients from the University Hospital of Dijon, with expert segmentations of the left-ventricular cavity (LV), myocardium (MYO) and right-ventricular cavity (RV) at end-diastole (ED) and end-systole (ES). Height and weight are also provided. Each patient belongs to one of five balanced groups (Table 1).

Table 1: ACDC diagnostic groups and the official train/test split.

Class	Meaning	Train (1–100)	Test (101–150)
NOR	Normal subjects	20	10
MINF	Systolic heart failure after myocardial infarction	20	10
DCM	Dilated cardiomyopathy	20	10
HCM	Hypertrophic cardiomyopathy	20	10
RV	Abnormal right ventricle	20	10

The official split is kept: patients 1–100 for training and model selection, patients 101–150 as an untouched test set used once at the end. All features are computed from the segmentations, so results reflect a setting where the masks are given (manual or from a segmentation network).

3 Feature extraction and reduction

Four published ACDC feature sets were merged into **656 features** and reduced to **313** by removing redundant ones in two stages. A clinical-only subset of **57 features** (the reduced set without Cetin radiomics) was also kept for comparison.

3.1 Feature sources

Table 2: Feature sources merged in this work.

Prefix	Source	Main content
W	Wolterink	Height, weight, ED/ES volumes of LV, RV, MYO, LV and RV ejection fraction, RV/LV and MYO/LV volume ratios
K	Khened	ED/ES volumes, EF, LV/RV and MYO/LV ratios, myocardial wall thickness (MWT) statistics per slice and across slices
I	Isensee	BSA-indexed volumes, myocardial mass, wall thickness (max, min, mean, std, septum), cavity circumference and circularity, volume-curve dynamics over the cardiac cycle
C	Cetin	PyRadiomics features per structure and phase: 3D shape, first-order intensity, GLCM, GLRLM, GLSZM texture, ED–ES duration

The four files were joined on patient id, after checking that the diagnostic labels agree for all 150 patients. Every column is prefixed with its source (W__, K__, I__, C__) so that its origin stays visible in the learned rules.

3.2 Stage 1 – physiological rules

When several sources measure the same physiological quantity, one representative was kept. Features that are exact mathematical functions of kept features were also dropped (Table 3).

Table 3: Stage-1 redundancy rules.

Group	Dropped	Kept instead	Reason
Anthropometrics	K, I, C copies of height and weight; I BMI	W height, W weight	Identical copies; BMI is a function of height and weight
Chamber and myocardial volumes	W, K, C (mesh and voxel) raw volumes of LV, RV, MYO at ED and ES; I myocardial mass; I volume-curve max and min	I BSA-indexed volumes	Same quantity; indexed volume is the clinical standard (ESC/ASE); mass = volume $\times$ 1.05 g/ml; curve max/min = ED/ES volume
Ejection fraction	K EF; I dynamic EF (LV, RV)	W EF_LV, EF_RV	Duplicates. EF is kept despite being computable from volumes: it encodes systolic function, and a TM cannot form ratios from thresholds
Volume ratios	K LV/RV and MYO/LV; I ES ratios	W RV/LV, MYO/LV (ED, ES)	LV/RV = 1/(RV/LV); RV/MYO is a function of kept ratios
Wall thickness	K max_mean and std_mean of MWT	I max and std thickness	Same quantities. K across-slice std (apex-to-base heterogeneity) is new information and kept
Cetin shape	Compactness1/2, SphericalDisproportion, SurfaceVolumeRatio, Minor/LeastAxisLength, Maximum2DDiameter Row/Column	Sphericity, SurfaceArea, Elongation, Flatness, Maximum3DDiameter	Exact functions (e.g. Compactness2 = Sphericity ³)
Cetin first-order and GLCM	TotalEnergy, Variance, Range, RootMeanSquared, GLCM SumAverage	Energy, StandardDeviation, Max, Min, Mean, JointAverage	Exact identities (e.g. Variance = SD ² , SumAverage = 2 $\times$ JointAverage)

3.3 Stage 2 – data-driven check (training patients only)

1. Features constant on patients 1–100 were removed.
2. Spearman correlation was computed between all survivors on the **training set only**, so no test information enters the selection.
3. Features were visited in order of clinical priority: clinical/physiological (W, K, I) first, then Cetin shape and timing, then first-order intensity, then texture (GLCM, GLRLM, GLSZM).
4. A feature with $|\rho| \geq 0.95$ to an already kept feature was dropped in favour of that feature.

Every original feature, its decision, its reason and the feature kept in its place are logged in `feature_reduction_report.csv`. The two outputs used for modelling are `merged_features_reduced.csv` (313 features) and `merged_features_all.csv` (656 features).

4 Model architecture and pipeline

The classifier is a multi-class Tsetlin Machine (`pyTsetlinMachine.MultiClassTsetlinMachine`) on thermometer-encoded features, tuned by 5-fold stratified cross-validation on patients 1–100 and tested once on patients 101–150 (Figure 1).

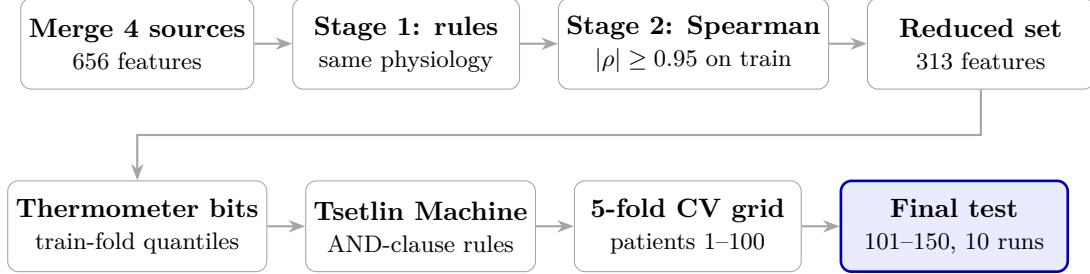


Figure 1: Full pipeline: features are reduced on training data, the TM is tuned by cross-validation and tested once on patients 101–150.

4.1 Thermometer encoding

A TM works on binary inputs. Each feature is turned into $n_{\text{bins}} - 1$ bits, where

$$b_k = \mathbf{1}[x \geq q_k], \quad k = 1, \dots, n_{\text{bins}} - 1, \quad (1)$$

and q_k are the quantiles of that feature on the training data (duplicate thresholds are merged). The thresholds are fitted on the training part of each fold only, so validation and test patients never influence the encoding. With 5 bins the 313 reduced features become 1247 bits.

4.2 Tsetlin Machine

For each of the 5 classes the TM holds a team of clauses. Each clause is a conjunction (AND) of literals, i.e. input bits or their negations, chosen by Tsetlin automata during training. Half of the clauses vote for the class (positive polarity), half against it. The class score is the sum of votes, clipped to $\pm T$, and the prediction is the class with the highest score:

$$\hat{y} = \arg \max_c \text{clip} \left(\sum_j C_j^{+,c}(\mathbf{b}) - \sum_j C_j^{-,c}(\mathbf{b}), -T, T \right). \quad (2)$$

Three hyperparameters control the model: the number of clauses, the vote threshold T (set to clauses/4) and the specificity s (higher s pushes clauses to include more literals). Because each literal is “feature $\geq$ threshold” or “feature $<$ threshold”, every clause reads as a clinical rule.

4.3 Training protocol

1. **Grid search:** $n_{\text{bins}} \in \{5, 10\}$, clauses $\in \{200, 500\}$, $s \in \{3, 8\}$, i.e. 8 configurations.
2. **5-fold stratified CV** (seed 42) in patients 1–100; each configuration is trained up to 40 epochs, and the validation accuracy is recorded after each epoch.
3. **Selection:** the configuration and epoch with the highest mean validation accuracy over the folds (earliest epoch on ties).
4. **Final model:** retrained on all 100 training patients with the selected settings and evaluated on patients 101–150. Since the TM is stochastic, this is repeated 10 times and mean $\pm$ std is reported; the median run is used for the confusion matrix and the extracted rules.

The same protocol is run on three feature sets: **reduced** (313 features), **clinical** (57 features: W, K, I only) and **all** (656 features, no reduction).

5 Results

The clinical-only model is best on every measure: 96.0% CV accuracy and $90.8\% \pm 2.7\%$ test accuracy, against about 84% for the two sets that include radiomics.

5.1 Overall accuracy

Table 4: Selected configuration and accuracy per feature set. Config = bins / clauses / T / s / epochs.

Feature set	Features	Bits	Config	CV acc. (1–100)	Test acc. (101–150)	Test range
Clinical (W+K+I)	57	491	10/500/125/3/16	0.960 ± 0.020	0.908 ± 0.027	0.86–0.94
Reduced (all)	313	1247	5/200/50/3/17	0.920 ± 0.081	0.844 ± 0.029	0.78–0.90
All (no reduction)	656	5873	10/200/50/3/32	0.920 ± 0.068	0.836 ± 0.028	0.78–0.88

All three sets selected $s = 3$; every $s = 8$ configuration learned more slowly and plateaued 10–20 points lower in CV. Redundancy removal cut the input size by $4.7\times$ ($5873 \rightarrow 1247$ bits) and halved training time ($406\text{ s} \rightarrow 183\text{ s}$ for the CV grid) at equal accuracy. Removing the radiomics as well gained about 6 points on the test set.

5.2 Confusion matrices and per-class metrics

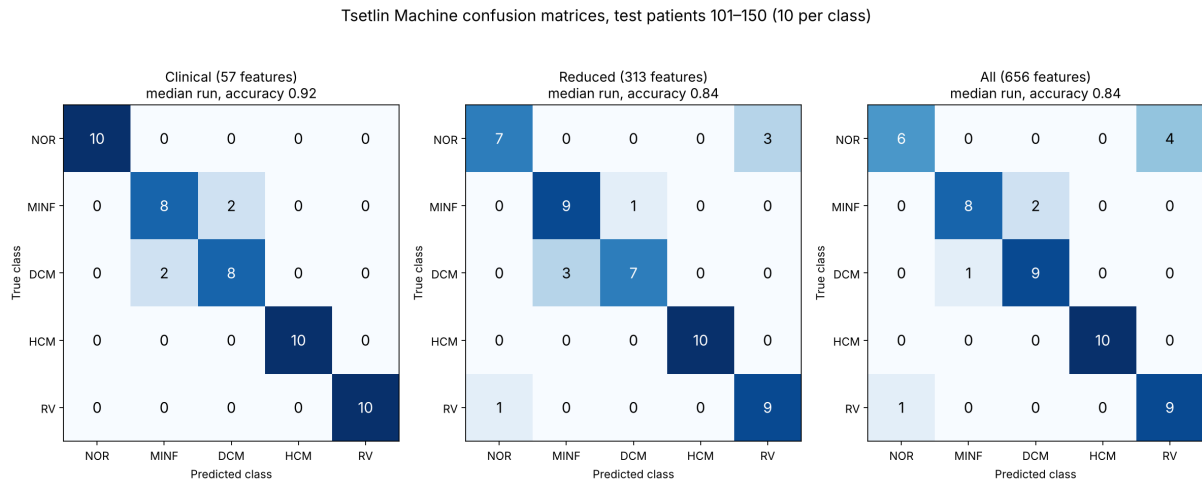


Figure 2: Confusion matrices of the median of the 10 final runs for the three feature sets; each row sums to the 10 test patients of that class.

Table 5: Per-class F1 score on the test set (median run).

Class	Clinical	Reduced	All
NOR	1.000	0.778	0.706
MINF	0.800	0.818	0.842
DCM	0.800	0.778	0.857
HCM	1.000	1.000	1.000
RV	1.000	0.818	0.783
Macro average	0.920	0.838	0.838

- **HCM** is recognised perfectly by every model: wall thickness separates it cleanly.
- **MINF** $\leftrightarrow$ **DCM** is the only confusion of the clinical model (2 + 2 patients). Both classes have a dilated LV with low ejection fraction; what separates them (regional wall motion and scar) is not captured by global volumes and thickness.
- **NOR** $\rightarrow$ **RV** appears only when radiomics are added (3 of 10 normals in the reduced set, 4 in the full set). The texture features seem to add noise that pushes normal hearts towards the RV class.

5.3 Calibration

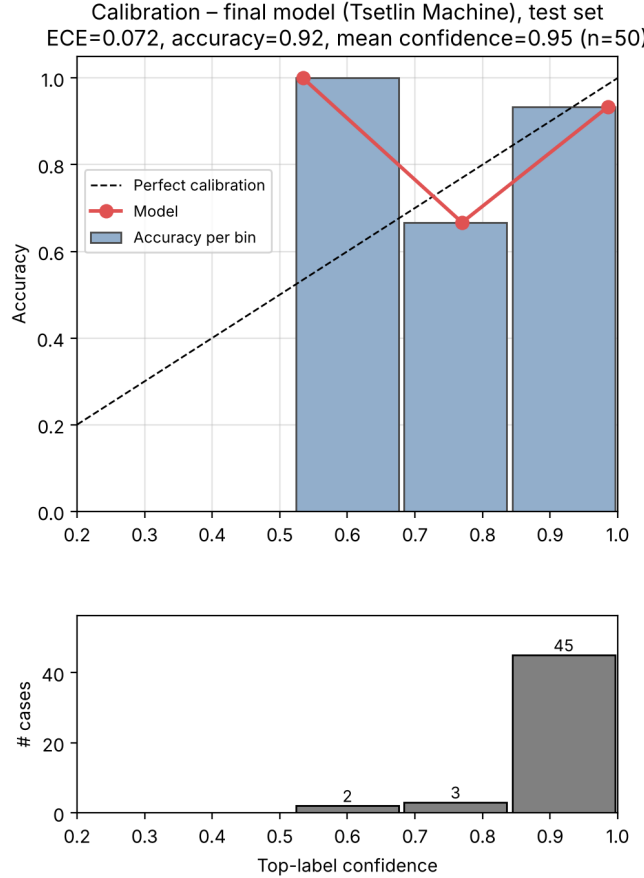


Figure 3: Reliability diagram of the final Tsetlin Machine (clinical feature set) on the test set.

The final TM is well calibrated overall: expected calibration error (ECE) 0.072, accuracy 0.92 and mean confidence 0.95, so it is only slightly over-confident (+0.03). Its confidences are

concentrated at the top: 45 of 50 test patients have confidence above 0.85, with an observed accuracy of 0.93 in that bin. Three of its four errors fall in this high-confidence bin; only one error has confidence below 0.85.

5.4 Calibration compared with the other models

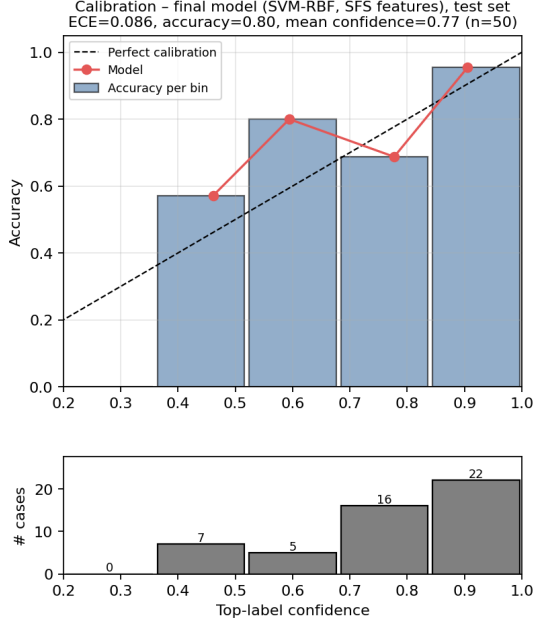
The TM has the lowest ECE of the five final models and is the only over-confident one; the SVM, the Wolterink random forest, the Khened two-stage classifier and the MLP+RF ensemble are all under-confident (Figure 4). ECE: TM 0.072, SVM 0.086, Wolterink RF 0.095, Khened 0.126, MLP+RF 0.145.

- **Lowest ECE:** the TM (0.072), followed by the SVM (0.086) and the Wolterink RF (0.095). The SVM’s confidences spread over the whole range and roughly follow the diagonal, but its accuracy is the lowest (0.80).
- **Under-confidence:** the Wolterink RF, Khened and MLP+RF report confidences 7–9 points below their accuracy (RF 0.78 vs 0.86, Khened 0.87 vs 0.94, MLP+RF 0.75 vs 0.84). Above 0.7 confidence they are right 86–100% of the time, so their probabilities are pessimistic rather than misleading.
- **Confidence of the TM:** the TM is the most confident model (mean 0.95) and also among the most accurate (0.92). Its calibration error is small, but because almost every patient gets confidence above 0.85, its confidence separates correct from wrong predictions poorly: 3 of its 4 errors are high-confidence.
- **Usefulness for triage:** the Khened classifier separates errors best: all three of its errors have confidence below 0.68, so a confidence threshold of 0.7 would send every misclassified patient to manual review (5 of 50 patients flagged). For the TM, the same threshold would flag 2 patients and catch none of its errors.

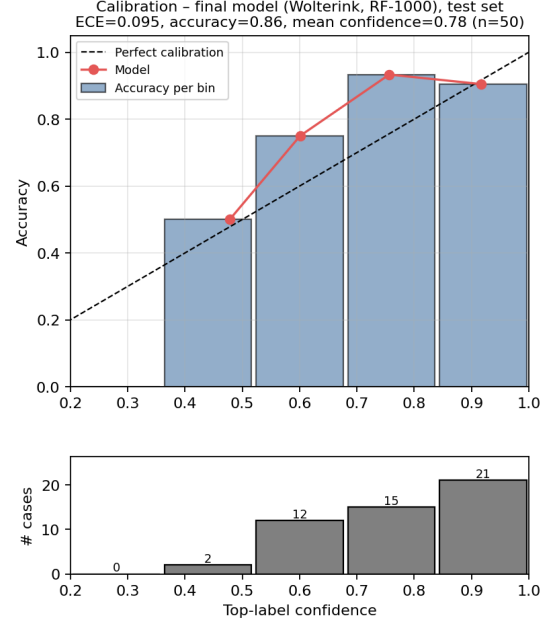
6 Conclusion

A Tsetlin Machine on 57 de-duplicated clinical features classifies the five ACDC pathologies with $90.8\% \pm 2.7\%$ test accuracy while producing rules a cardiologist can read.

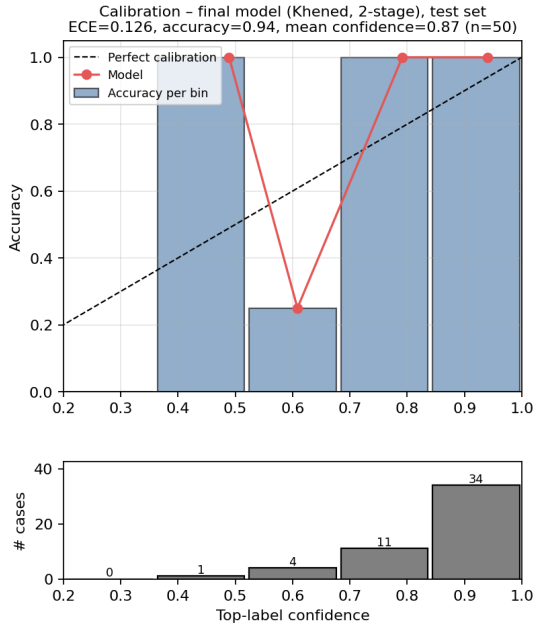
- **Physiology-driven feature reduction works.** Merging four feature sets and removing redundancy by rules plus a train-only correlation check cut the input from 656 to 313 features (5873 to 1247 bits) without loss of accuracy, and halved training time.
- **Fewer, meaningful features generalise better.** Dropping the radiomic texture features raised test accuracy from about 84% to 91% and removed the NOR $\rightarrow$ RV errors. With 100 training patients, hundreds of texture features mainly add noise.
- **The TM is the best-calibrated model.** It has the lowest ECE of the five final models (0.072 vs 0.086–0.145) and is only slightly over-confident, while the other models are under-confident. Its confidences are concentrated above 0.85, however, so they flag few of its errors; the Khened classifier places all its errors below 0.7 confidence.



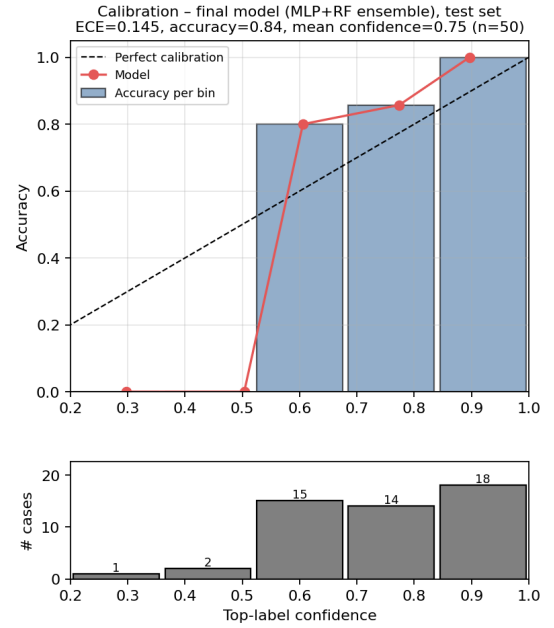
(a) Centin model



(b) Wolterink, RF-1000



(c) Khened, two-stage



(d) Isensee model, MLP + RF ensemble

Figure 4: Reliability diagrams of the other four final models on the test set (the TM is in Figure 3). Points above the diagonal are under-confident, points below are over-confident.